Student Information
Name: Jsh Agarwal UID: 2022200004 Assignment: DA9
- Aerosol Spectral Clustering Analysis

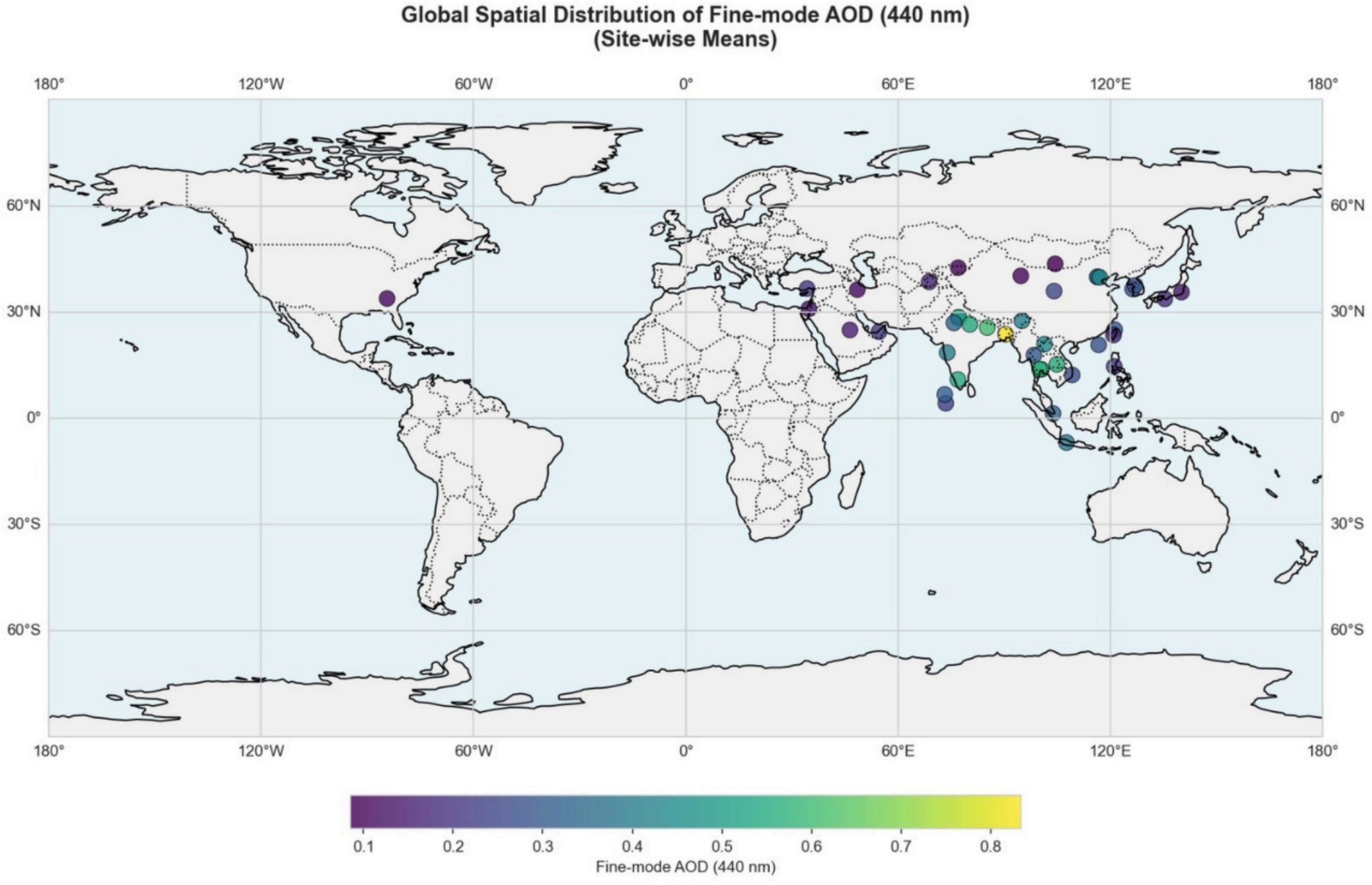
Aerosol Spectral Clustering Analysis Report

EXPERIMENT 9

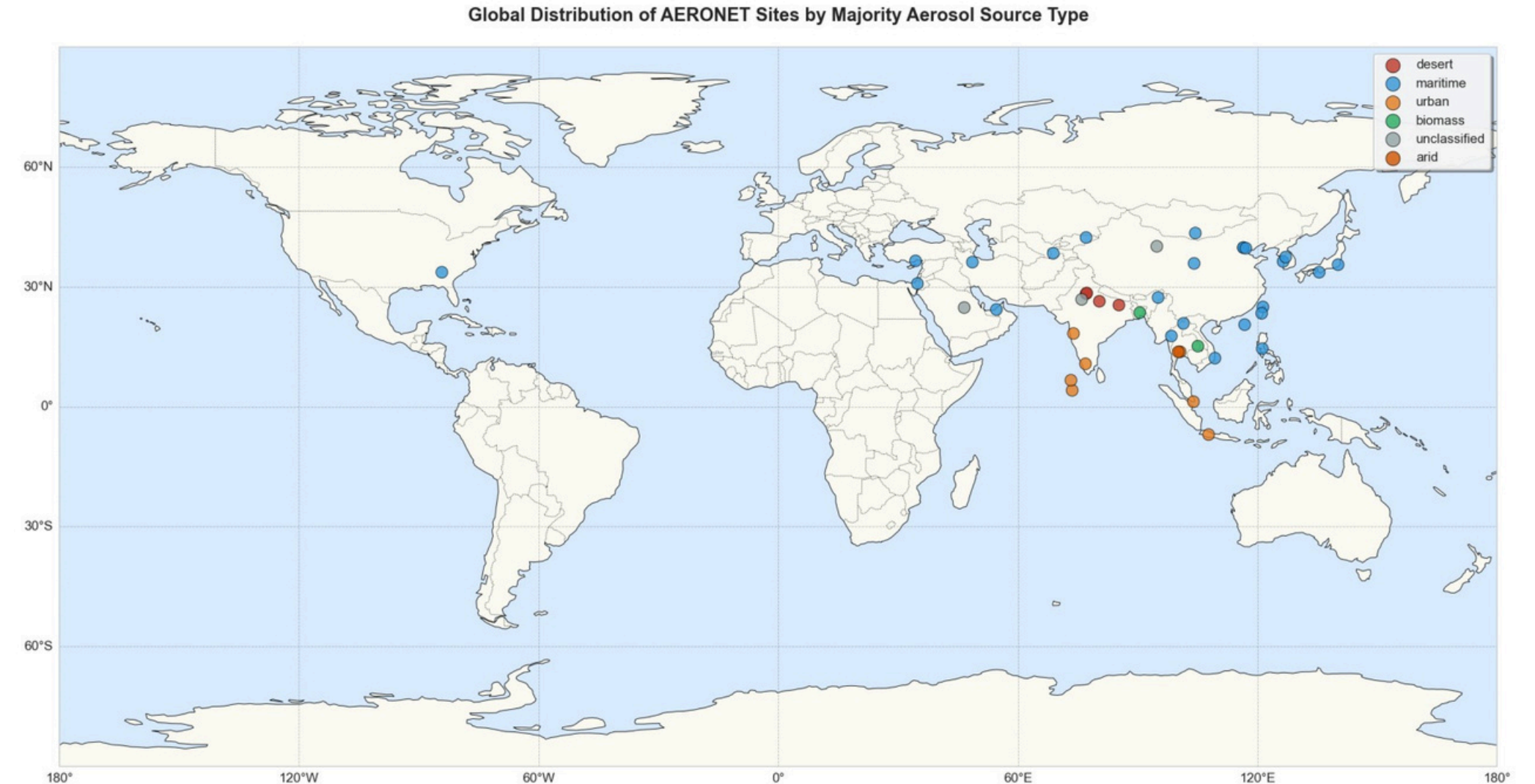
Following Ningombam et al., 2024 Methodology

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Global Spatial Distribution - Fine-mode AOD



Global Distribution by Source Type



Data Cleaning & Preprocessing Summary

DATA CLEANING SUMMARY
==========

Initial Dataset:

- Input file: combined_dataset.csv
- Initial rows: 186,562
- Initial columns: 48

Data Cleaning:
- Rows removed: 0 (0.00%)
- Final rows: 186,562
- Reason:Missingvalues in required columns

Required Columns:
- AOD_total(TotalAOD at 440 nm)
- EAE(ExtinctionAngstrom Exponent 440-870 nm)
- AAE(AbsorptionAngstrom Exponent 440-870 nm)
- SSA(SingleScattering Albedo at 440 nm)

Proxy Features Computed:

- AOD_fine:Directmeasurements used

- RRI:Directmeasurements used

Table 2: Aerosol Optical Parameters Statistics for ASIA Region

Parameter	Average	Median
AOD_T	0.40 ± 0.32	0.29
AOD_F	0.30 ± 0.29	0.19
EAE	1.06 ± 0.39	1.14
SSA	0.89 ± 0.03	0.88
Ref_R	1.48 ± 0.05	1.48
AAE	1.48 ± 0.36	1.47

Note: Values shown as Average ± Standard Deviation and Median
Based on AERONET Level 2.0 data for ASIA region sites

Feature Statistics (Before & After Standardization)

Original Feature Statistics

Feature	mean	std	min	25%	50%	75%	max
AOD_fine	0.3037	0.2893	0.0361	0.1083	0.1933	0.3844	1.5092
EAE	1.0599	0.3893	0.1436	0.7782	1.138	1.3638	1.7414
AAE	1.4786	0.359	0.6238	1.2267	1.4703	1.7162	2.7301
SSA	0.886	0.0303	0.8237	0.8629	0.8818	0.9065	0.9685
RRI	1.4774	0.0529	1.33	1.4402	1.4773	1.5124	1.6

Standardized Feature Statistics

Feature	mean	std	min	25%	50%	75%	max
AOD_fine	-0.0	1.0	-0.932	-0.677	-0.3799	0.2799	4.1724
EAE	-0.0	1.0	-2.3746	-0.7155	0.1901	0.7873	1.7501
AAE	-0.0	1.0	-2.4063	-0.7029	-0.0273	0.6636	3.4981
SSA	-0.0	1.0	-2.055	-0.7577	-0.1439	0.6722	2.7459
RRI	-0.0	1.0	-2.7785	-0.7048	0.0053	0.6606	2.3347

Table 3: Seasonal Distribution of Aerosol Properties (ASIA Region)

Season	AOD_Fine	AOD_Coarse	EAE	SSA	AAOD	AAE
Pre monsoon	0.32 ± 0.28	nan	0.96 ± 0.42	0.89 ± 0.03	nan	1.50 ± 0.37
DJF	0.35 ± 0.33	nan	1.20 ± 0.31	0.89 ± 0.03	nan	1.43 ± 0.35
Post monsoon	0.29 ± 0.29	nan	1.10 ± 0.34	0.89 ± 0.03	nan	1.48 ± 0.34
Monsoon	0.22 ± 0.20	nan	0.96 ± 0.42	0.89 ± 0.03	nan	1.52 ± 0.37

Note: DJF=Dec-Jan-Feb (Winter), MAM=Mar-Apr-May (Spring), JJA=Jun-Jul-Aug (Summer), SON=Sep-Oct-Nov (Autumn)

Table 4: Cluster Centroids with Aerosol Type Classification (ASIA)

Cluster	Percentage	Aerosol_Type	Fine_mode_AOD	EAE	AAE	SSA	RRI
1	73.9	M	0.18	0.97	1.56	0.88	1.48
2	26.1	B	0.66	1.33	1.25	0.92	1.48

Note: Aerosol Types: D=Dust, B=Biomass Burning, U=Urban/Industrial, M=Mixed
Classification based on EAE, AAE, SSA thresholds from Ningombam et al., 2024

Table 5: Seasonal Distribution by Clusters (ASIA Region)

Season	Cluster_1	Cluster_2
Pre monsoon	74%, M	26%, B
DJF	65%, M	35%, B
Post monsoon	74%, M	26%, B
Monsoon	86%, M	14%, B

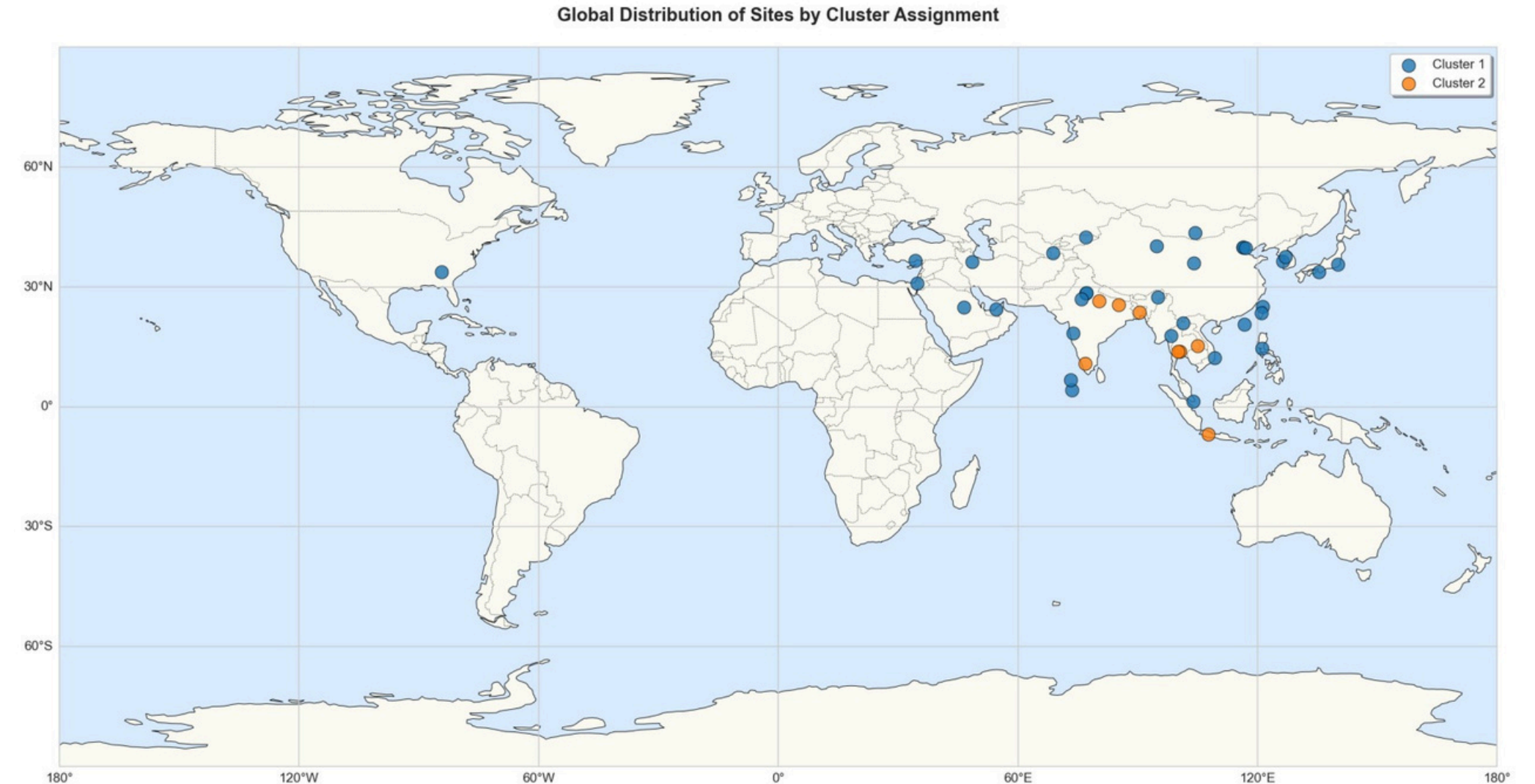
Note: Values show percentage and dominant aerosol type for each cluster
Format: Percentage%, AerosolType (D/B/U/M)

Table 6: Aerosol Classification Based on FMF and SSA (ASIA Region)

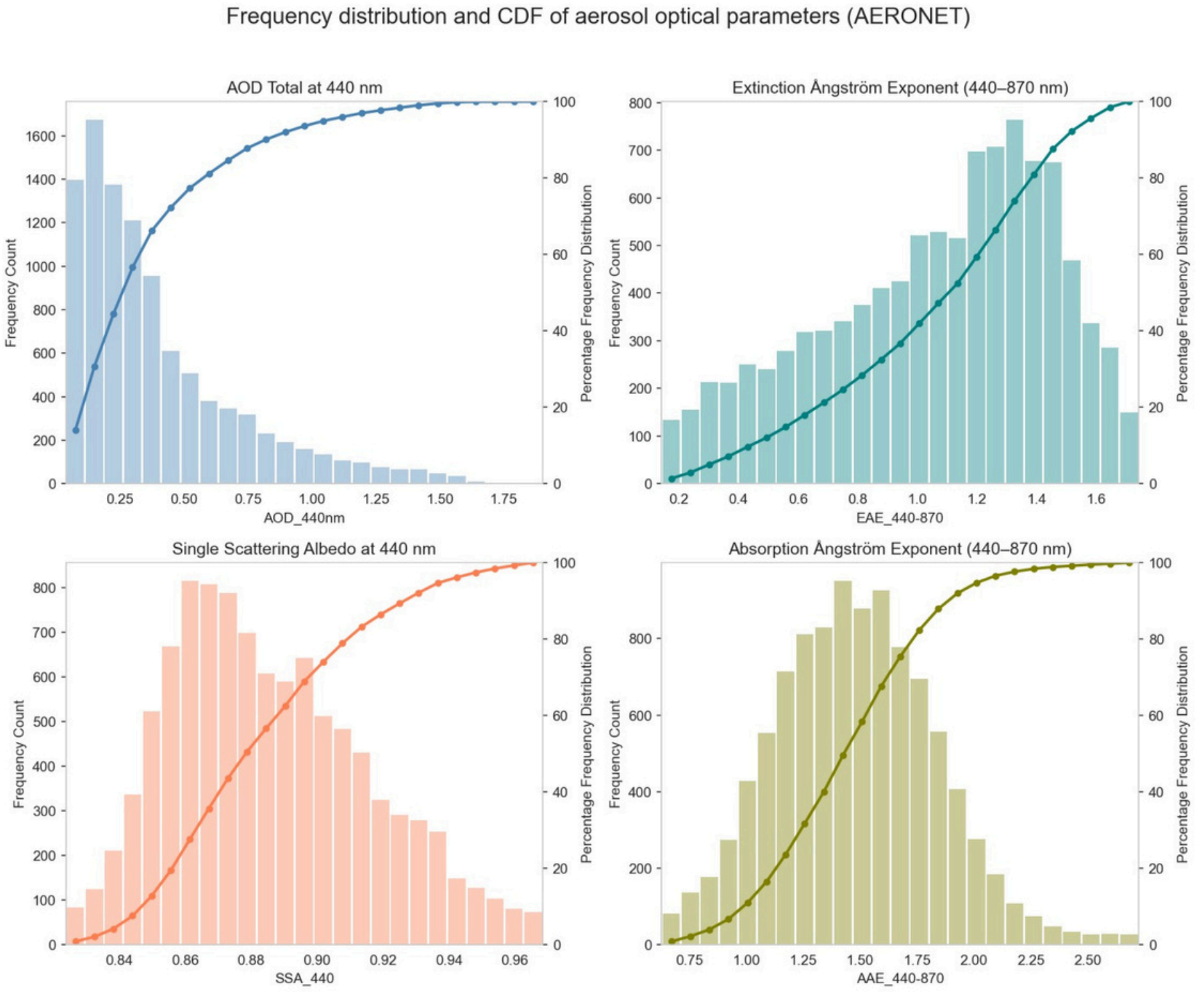
Uncer.	Dust	Mix.	N.A.	BC
0	21943	67681	4312	64541

Note: Classification criteria:
Uncer. = Uncertain/Unknown
Dust = FMF < 0.4 and SSA < 0.95
Mix. = Mixed aerosols (0.4 ≤ FMF < 0.6)
N.A. = Non-Absorbing (SSA > 0.95)
BC = Black Carbon (FMF ≥ 0.6 and SSA < 0.90)

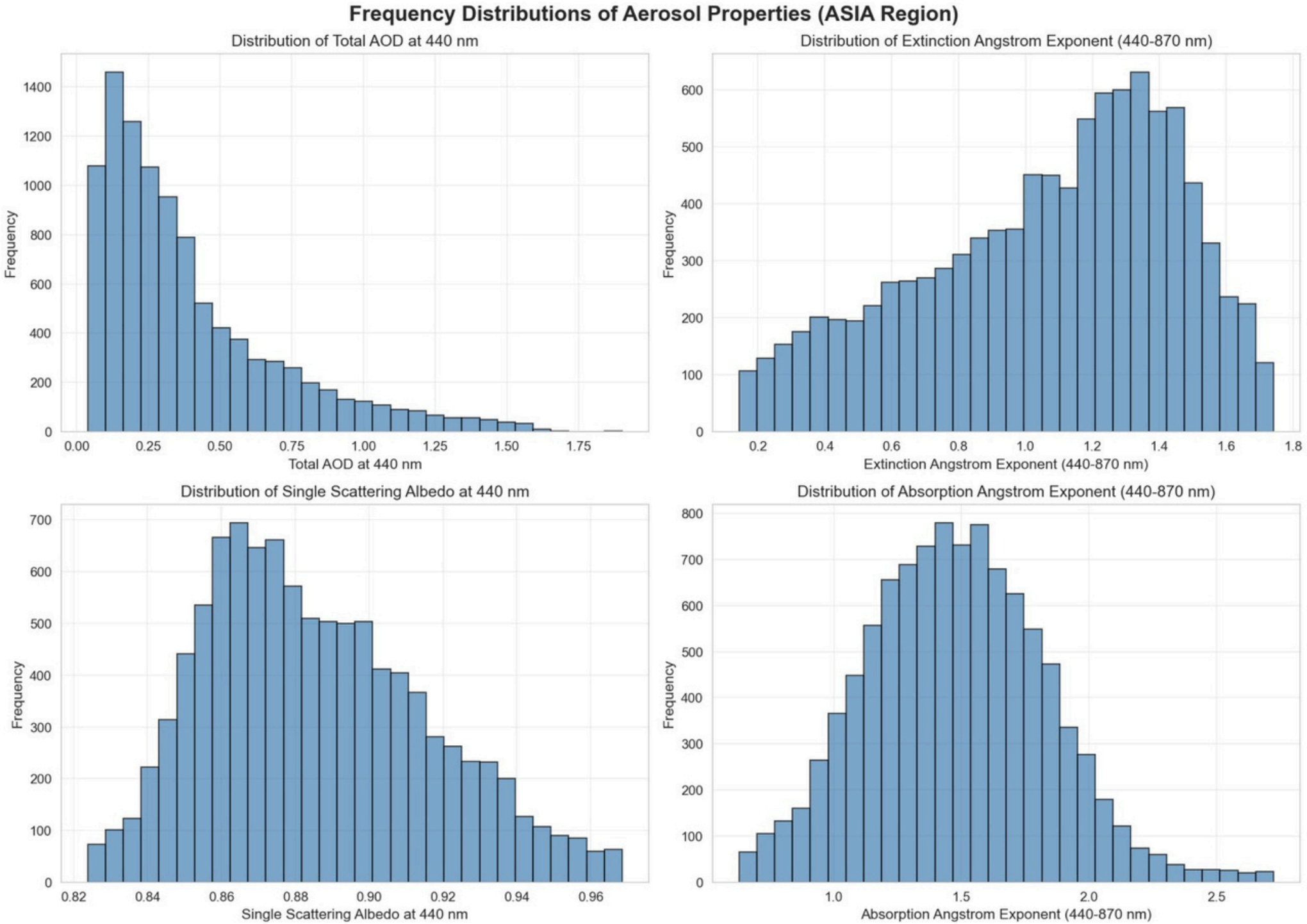
Global Distribution by Cluster Assignment



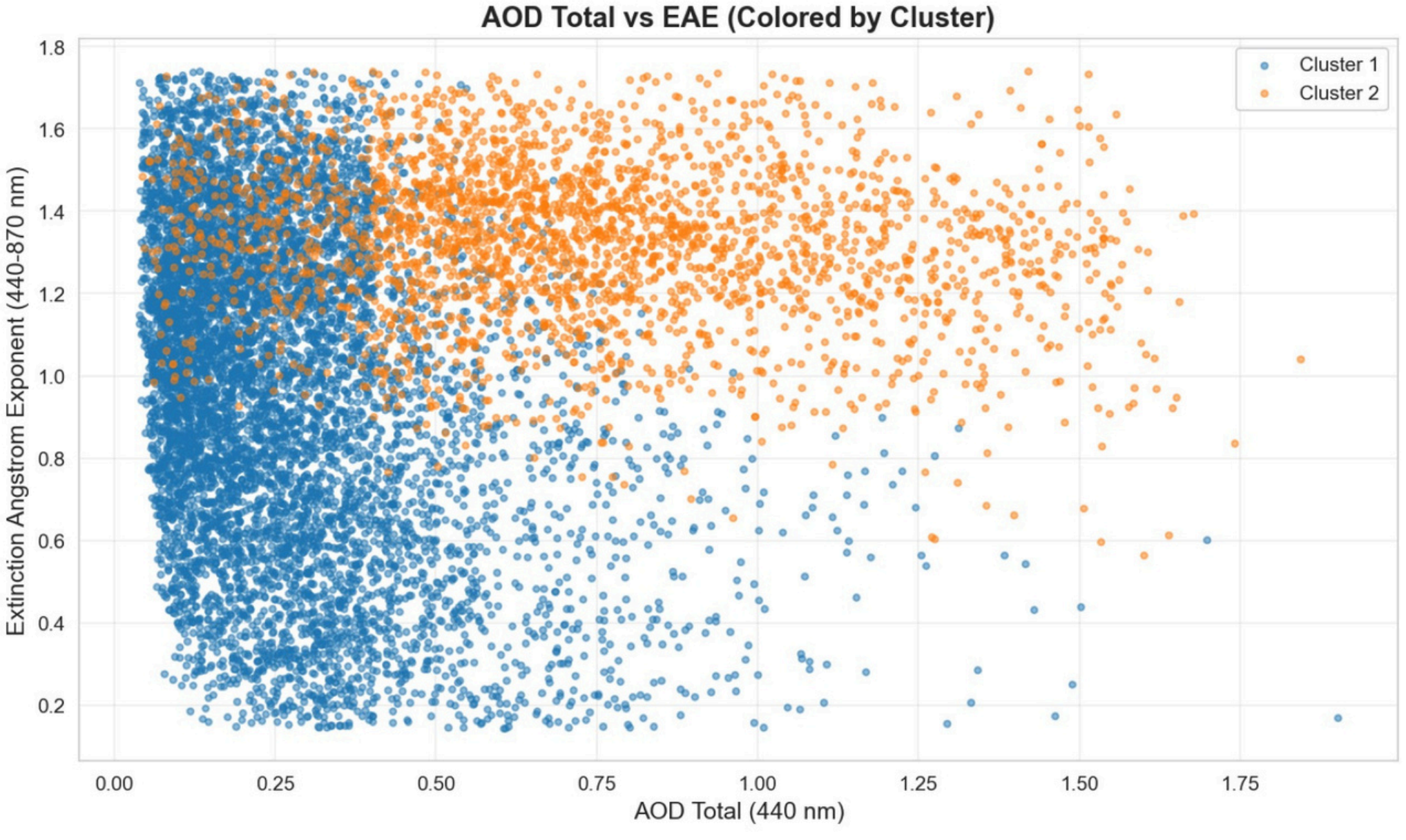
Frequency Distribution and CDF of Aerosol Parameters



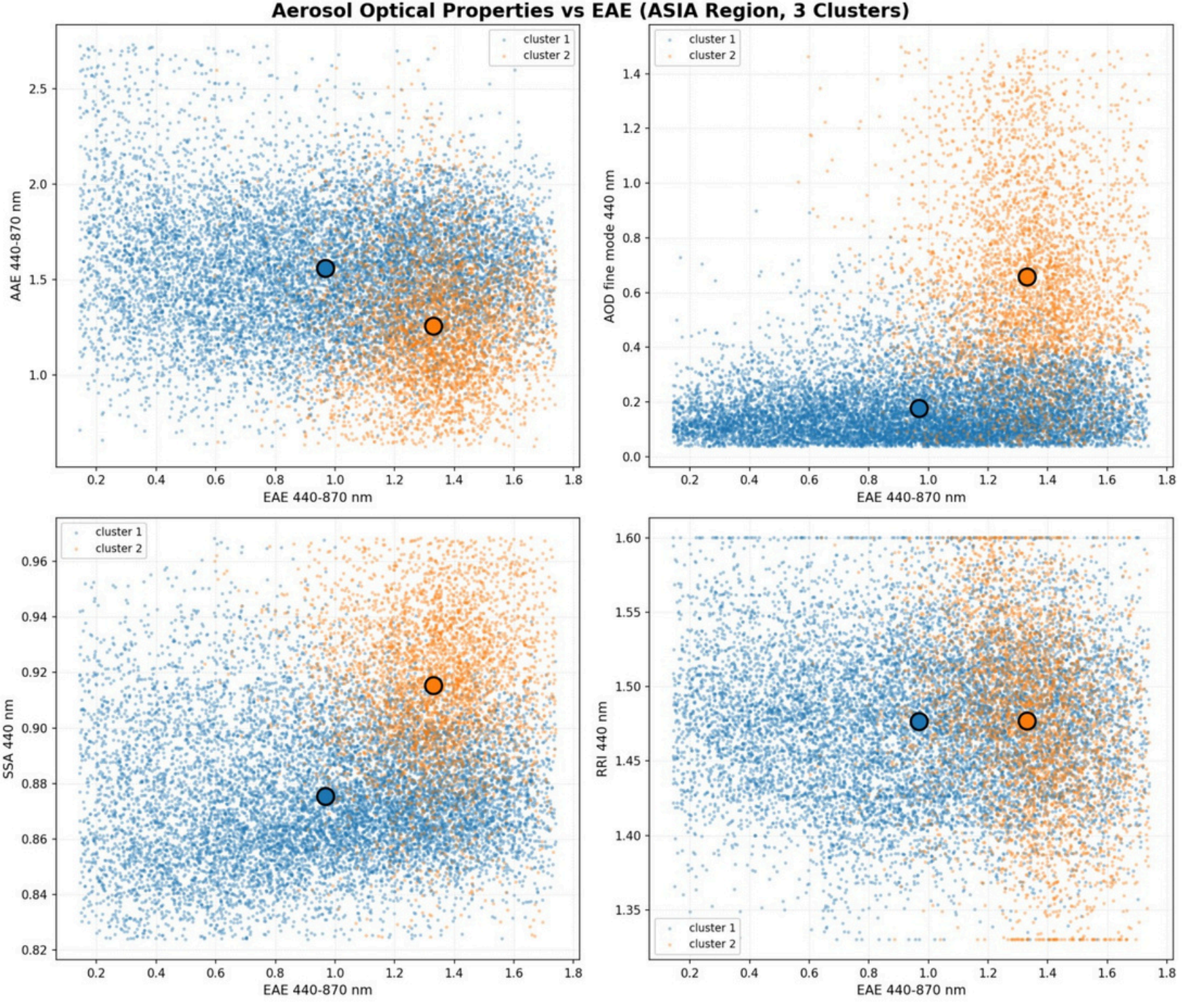
Frequency Distributions of Aerosol Properties (ASIA)



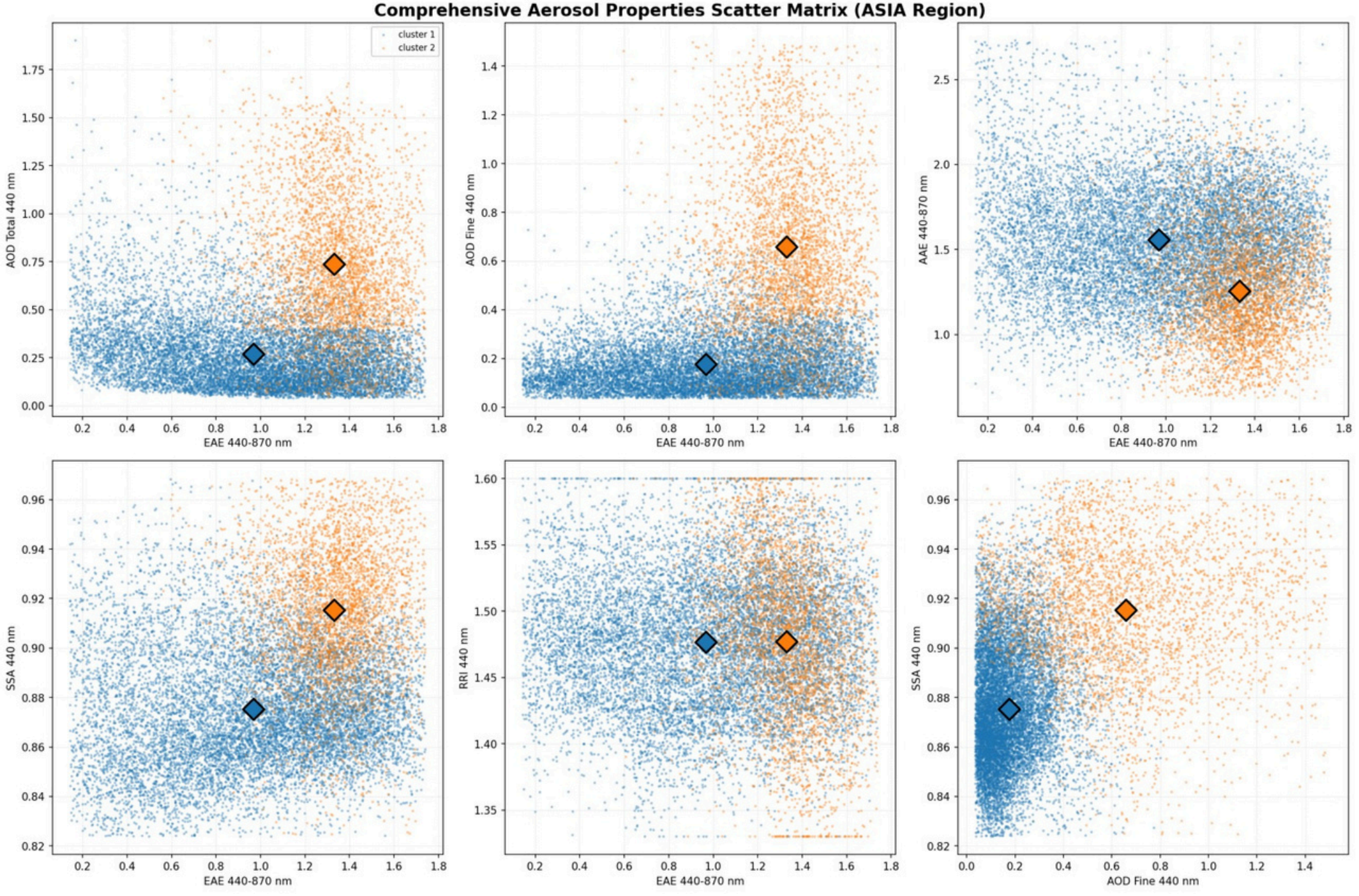
AOD vs EAE Scatter Plot



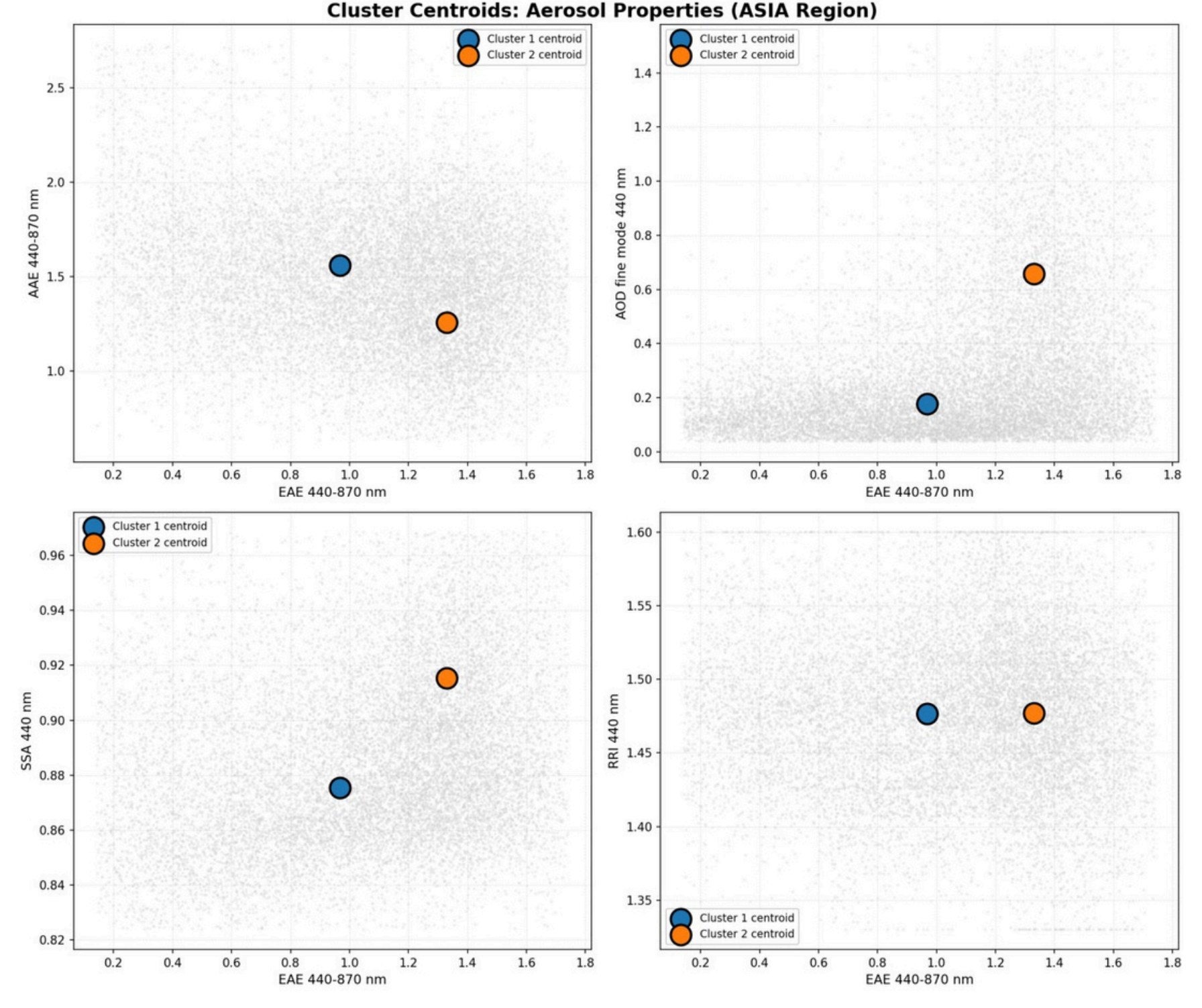
ASIA: 4-Panel Scatter Plots with Cluster Colors



ASIA: Extended 6-Panel Scatter Matrix

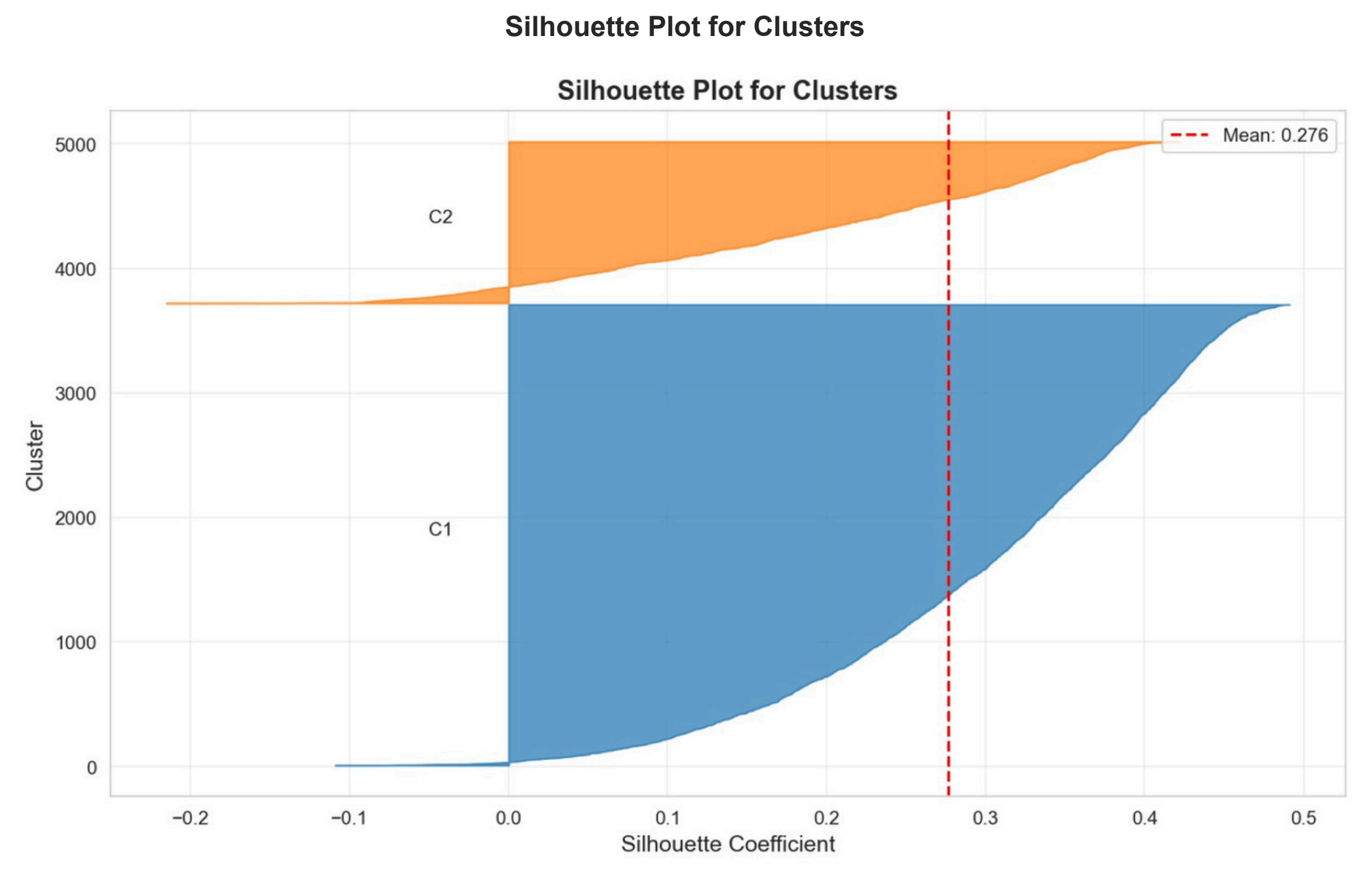


ASIA: Cluster Centroids Visualization

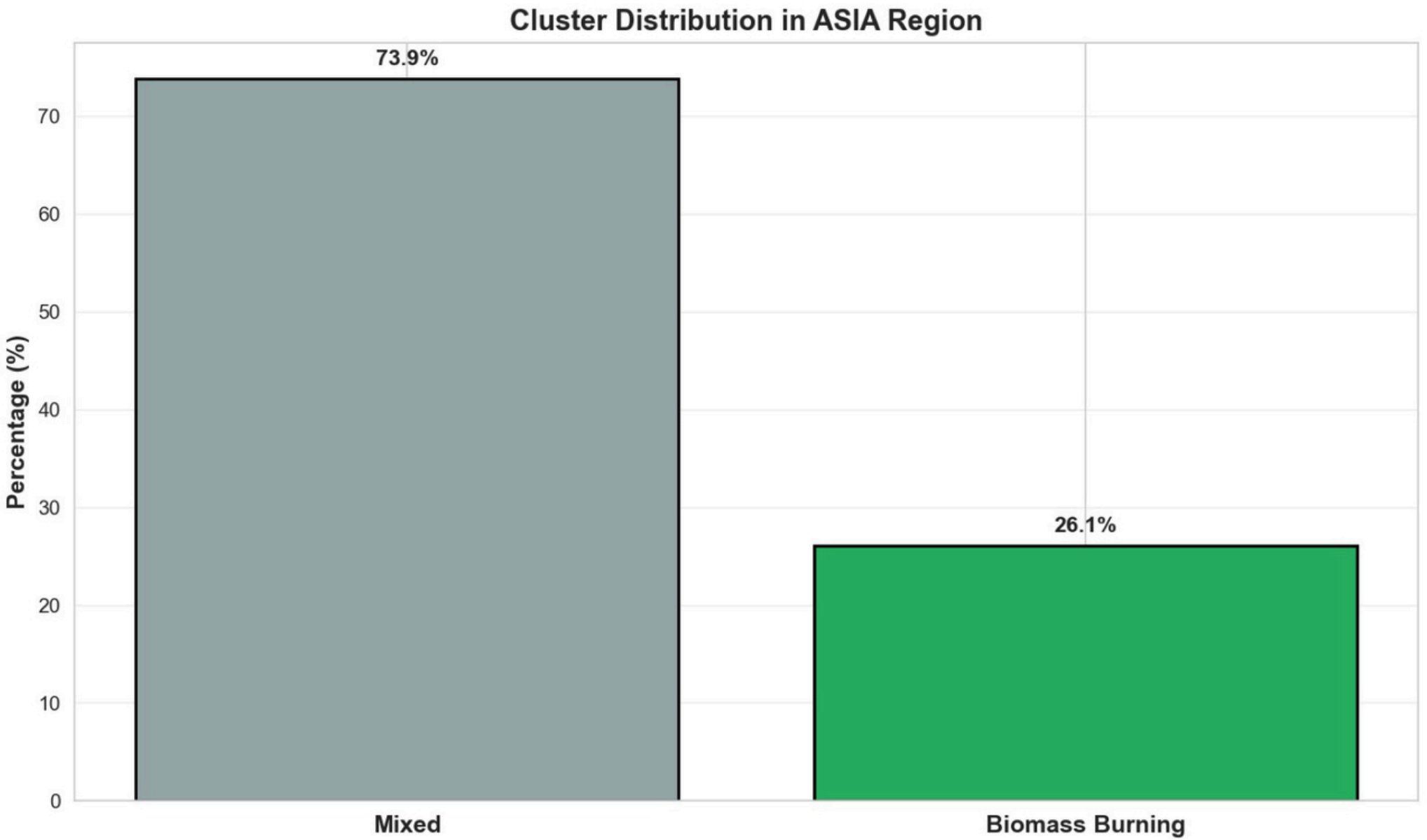


Aerosol Property Relationships (Colored by Cluster)




Silhouette Plot for Clusters
Silhouette Plot for Clusters
Mean: 0.276
C2
C1
Cluster
5000
4000
3000
2000
1000
0
−0.2
−0.1
0.0
0.1
0.2
0.3
0.4
0.5
Silhouette Coefficient

Cluster Distribution Bar Chart

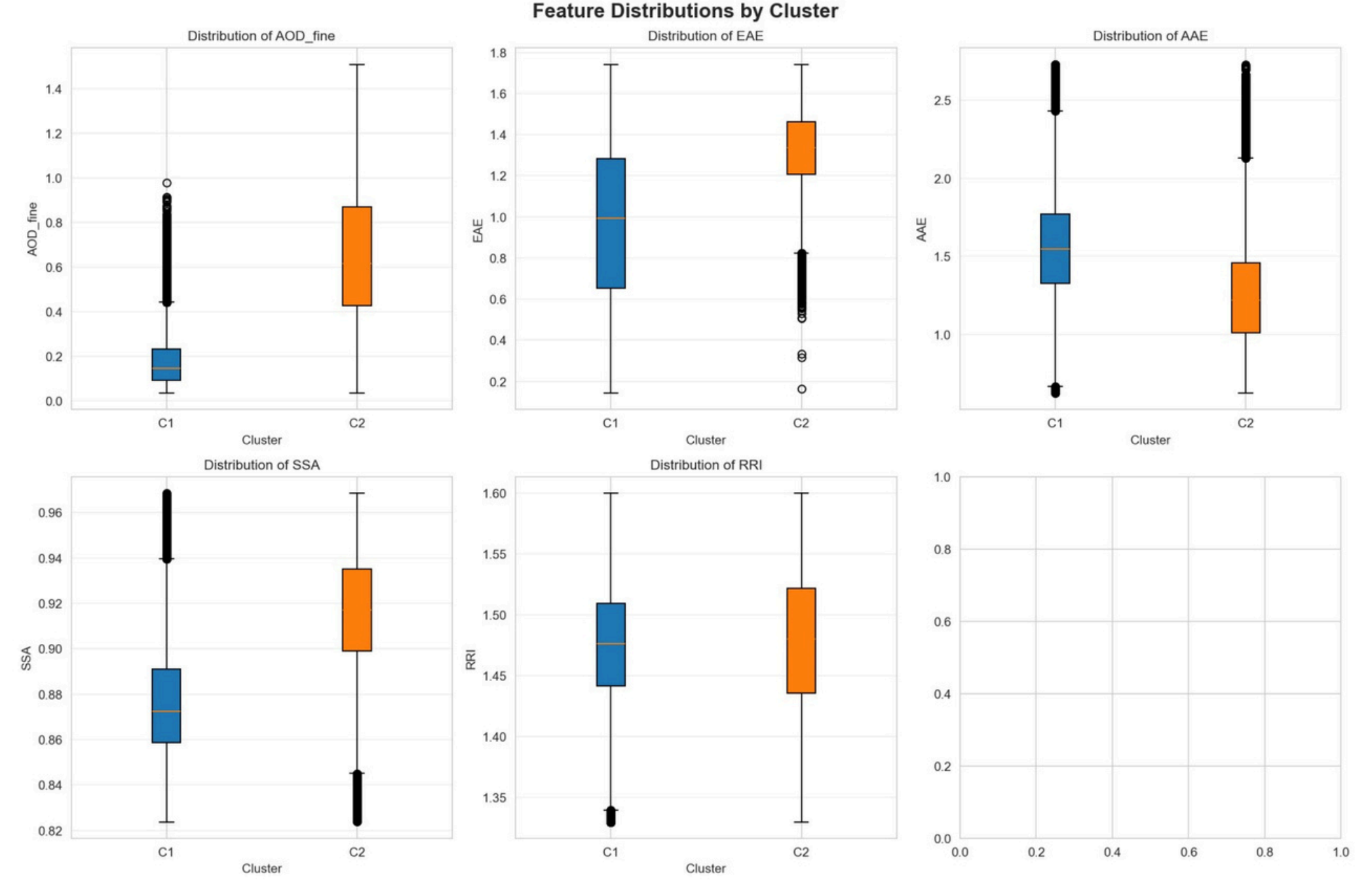


Cluster Centroids Table

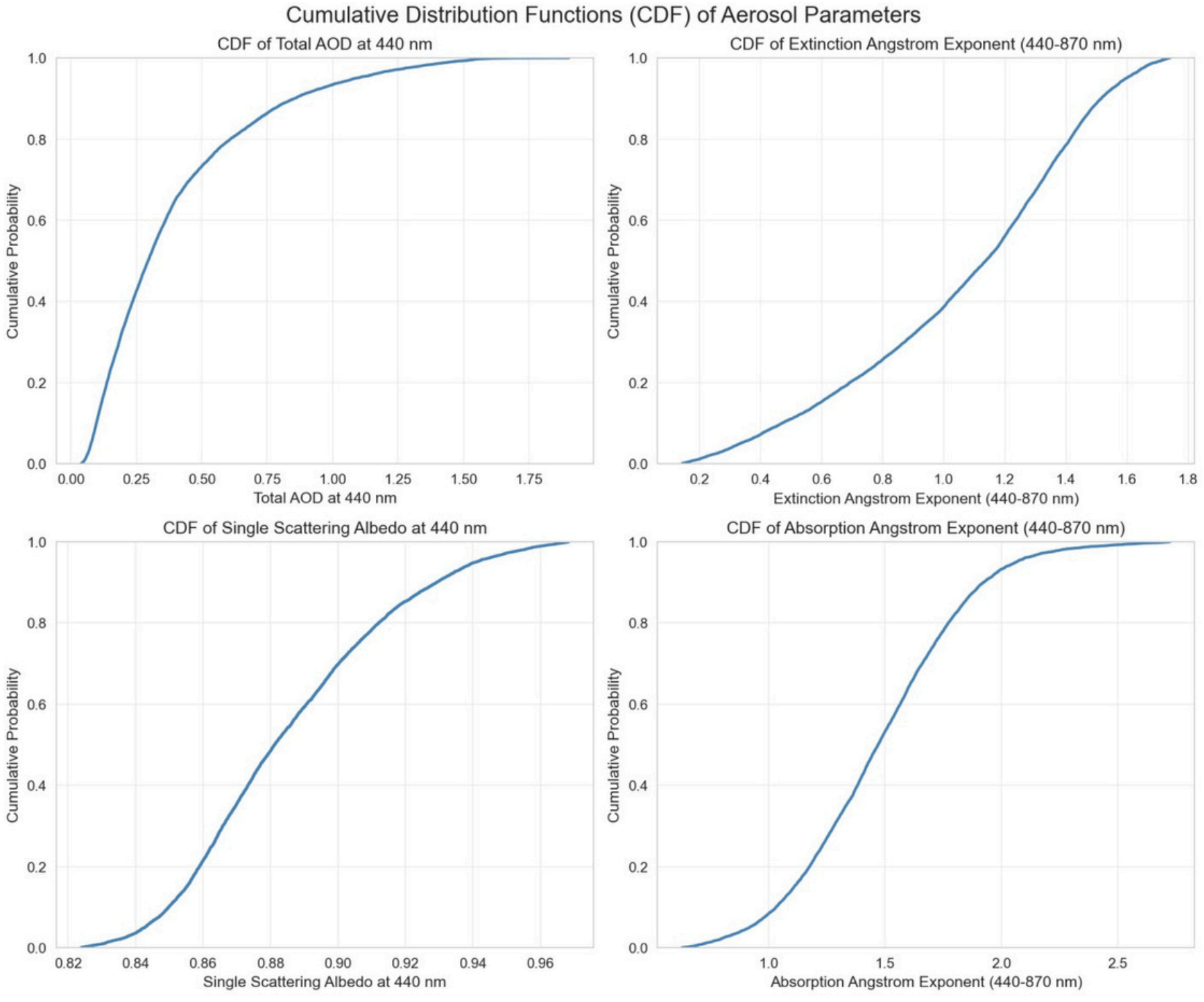
Cluster Centroids & Aerosol Type Classification

Cluster	Aerosol Type	AOD_fine	EAE	AAE	SSA	RRI
Cluster 0	Mixed	0.1763	0.9651	1.5584	0.8754	1.477
Cluster 1	Biomass Burning	0.664	1.3279	1.253	0.9159	1.4785

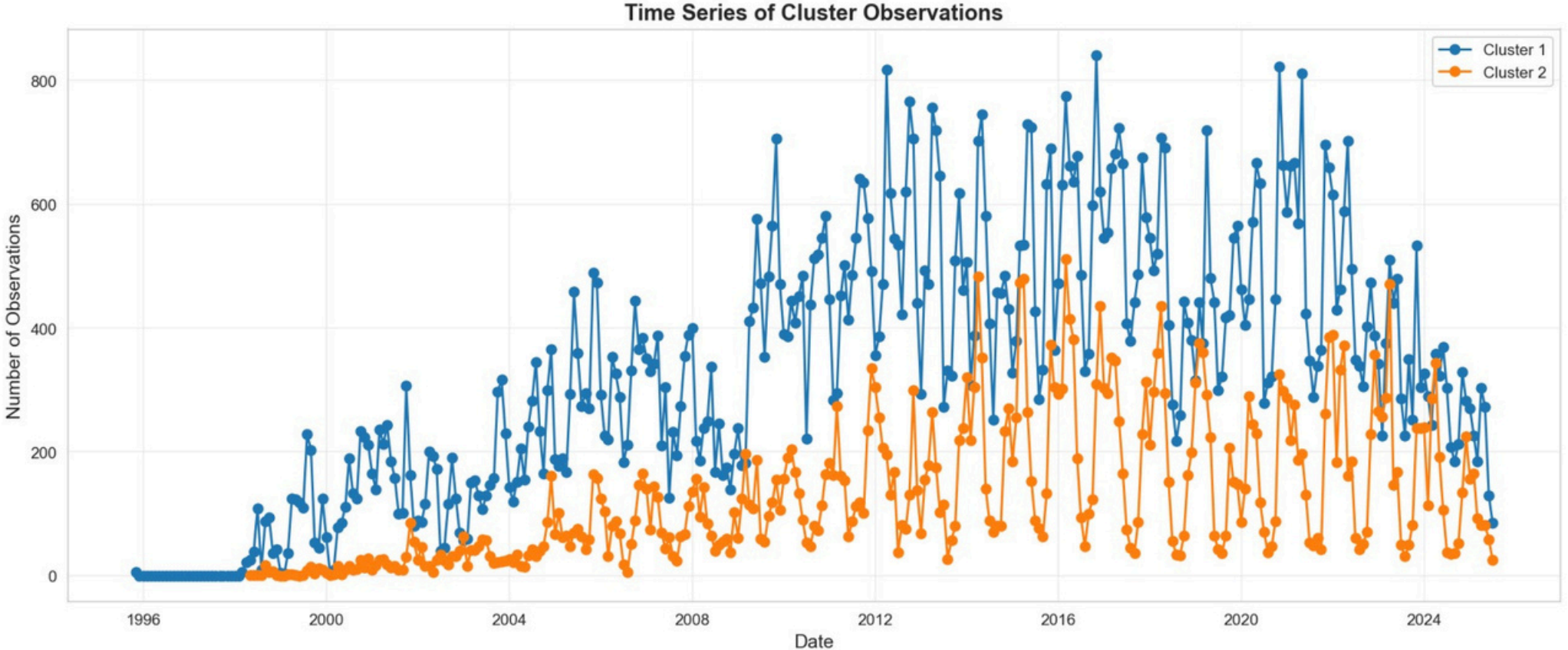
Feature Distributions by Cluster



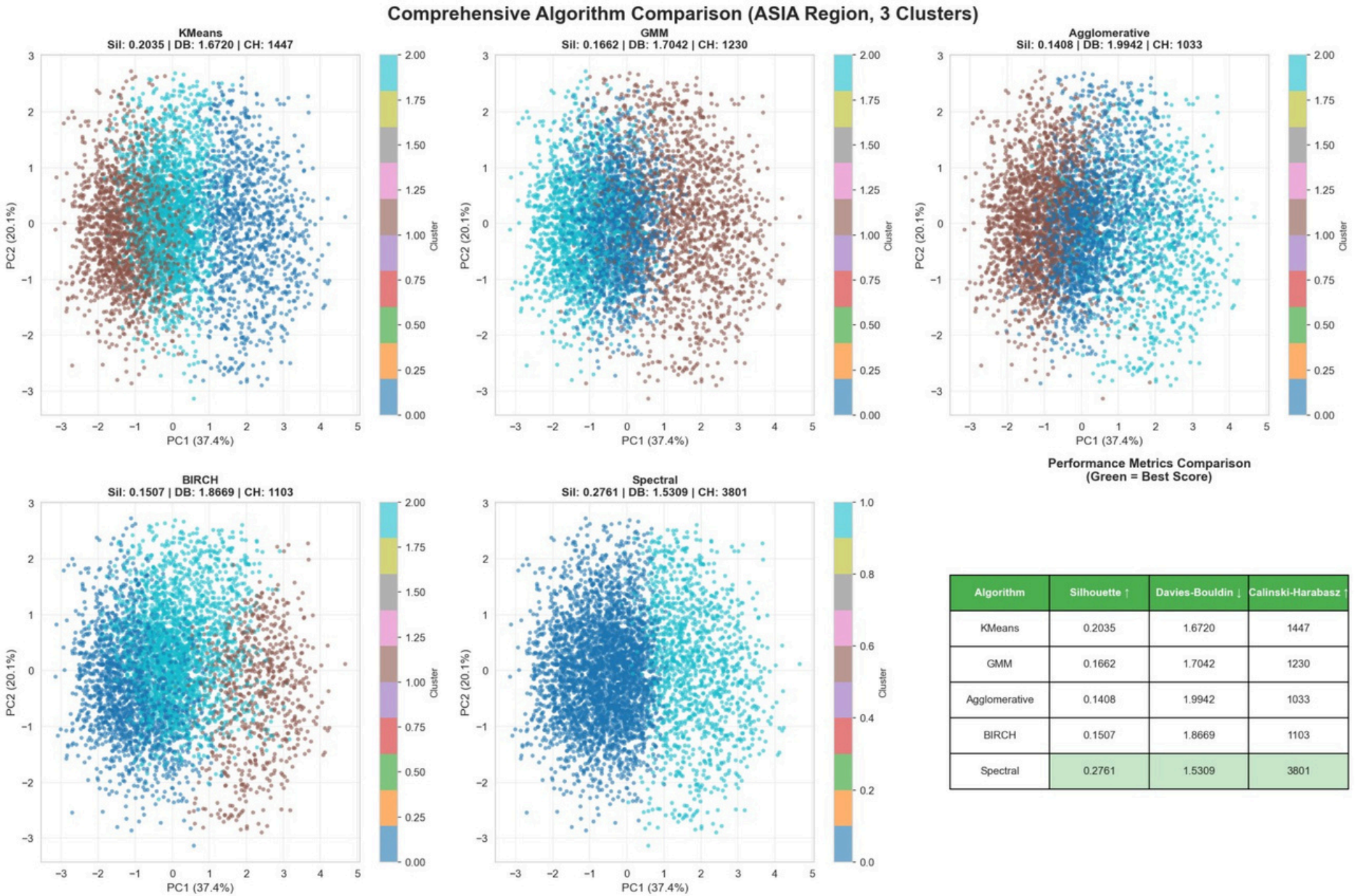
Cumulative Distribution Functions



Time Series of Cluster Observations



Algorithm Comparison: Spectral vs KMeans



Performance Metrics Comparison
(Green = Best Score)

Algorithm	Silhouette ↑	Davies-Bouldin ↓	Calinski-Harabasz ↑
KMeans	0.2035	1.6720	1447
GMM	0.1662	1.7042	1230
Agglomerative	0.1408	1.9942	1033
BIRCH	0.1507	1.8669	1103
Spectral	0.2761	1.5309	3801

Conclusions and Interpretation

1. Clustering Quality:

• Silhouette Score: 0.2761 Weak separation Algorithm:

•Spectral Clustering with 3 clusters Method: Follows

•Ningombam et al., 2024 methodology

2. Aerosol Composition (ASIA Region):

• Dominant type: Mixed (73.9%)

• Mixed: 73.9%

•Biomass Burning: 26.1%

3. Physical Interpretation:

Mixed:

- EAE=0.97, AAE=1.56, SSA=0.88 → Complex mixture

- AOD_fine: 0.1763, EAE: 0.965

- AAE: 1.558, SSA: 0.875

Biomass Burning:

- High EAE (1.33), Moderate AAE (1.25), Low SSA (0.92) → Absorbing fine particles

- AOD_fine: 0.6640, EAE: 1.328

- AAE: 1.253, SSA: 0.916
- Data rem oval: 0.00% due to missing values
- Total samples analyzed: 158,477

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5. Recommendations:

- Use this classification for source apportionment studies

- Consider seasonal variations in future analysis

- Validate results with meteorological data

- Apply to air quality management strategies

- Use cluster labels for predictive modeling

6. Key Findings:

- ASIA region shows mixed aerosol composition

- Mixed aerosols are predominant

- Clear spectral signatures distinguish aerosol types

- Clustering aligns with known aerosol physics

- Method suitable for large-scale aerosol classification